VICTORIA UNIVERSITY
Kampala · Uganda

EXAMINATION

FACULTY OF SCIENCE AND TECHNOLOGY

PROGRAMME(S): BBC, BAD

SESSION: DAY

MODULE CODE AND TITLE:	2101 ST - Probability And Statistics
LECTURER:	Dr. Ronald Bbosa

INSTRUCTIONS TO THE STUDENTS

1. V-class is the **ONLY** official submission platform.
2. You have **ONE WEEK** to complete this assignment. No submission will be accepted after one week (the system has closed).
3. Each group (maximum 5 members) should attempt one question that has been specifically assigned to them.
4. Read the business scenario carefully – Understand the question being asked.
5. Identify the appropriate statistical method – Based on the type of data, number of groups, and research question, decide which inferential technique to use.
6. Justify your choice – Explain in plain language why this method is correct for this situation.
7. Check all necessary assumptions – Perform and report the required assumption tests (normality, homogeneity of variance, independence, expected frequencies, etc.). Explain what each assumption means.
8. Run the analysis using JASP or Jamovi
9. Interpret the results – Translate p-values, confidence intervals, effect sizes, etc., into business language that a manager would understand.
10. State your conclusion and recommendation – Clearly say whether the null hypothesis is rejected or not, and provide actionable advice to the organisation.
11. Reflect on limitations – What could affect the validity of your conclusion? What would you do differently with more data or resources?
12. Review your answers carefully before submitting your assignment.
13. Each group should submit one comprehensive report showing all work, clearly stating and explaining concepts used. Include screenshots, project files. The submission deadline is **Sunday 21st -06-2026 11:59PM.**
14. In each submission, the group number should be indicated alongside the details (name and student ID) of all the members in that group.

Each group should attempt one question

QUESTION ONE (Group #1)

An airline company is evaluating customer satisfaction across different travel classes and customer types. They have collected survey responses from over 120,000 passengers. The marketing team wants to:

a) Verify whether the average satisfaction rating (on a 0–5 scale for various service attributes) meets the company target of 4.0.

b) Compare satisfaction scores between different customer types (Loyal vs Disloyal customers).

c) Assess whether satisfaction differs between Business Class and Economy Class passengers for the same flight routes.

Use the dataset below to perform the following statistical tasks:

Dataset: Airline Passenger Satisfaction Dataset – Use:

- Satisfaction (overall satisfaction: Satisfied, Neutral, Dissatisfied)
- Customer Type (Loyal customer, Disloyal customer)
- Class (Business, Eco, Eco Plus)
- Inflight service (satisfaction rating 0–5)
- Seat comfort (satisfaction rating 0–5)
- Food and drink (satisfaction rating 0–5)
- Cleanliness (satisfaction rating 0–5)

Dataset Link: https://www.kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction

Part A: The airline claims that the average inflight service satisfaction rating is 4.0 out of 5. The customer service director believes the true average is lower than 4.0..

1. State the null and alternative hypotheses in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Perform the appropriate test in JASP/Jamovi to test whether the mean inflight service rating is less than 4.0. Report: test statistic, df, p-value, and 95% CI.
3. Interpret the results and state your conclusion.

Part B: The airline wants to compare inflight service satisfaction ratings between Loyal customers and Disloyal customers.

1. State the null and alternative hypotheses for testing whether there is a difference in satisfaction between Loyal and Disloyal customers in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Check and interpret the assumptions.
3. Run the appropriate test and report the test statistic, df, p-value, and 95% CI for the difference.
4. Interpret the results and provide a recommendation.

Part C: The airline wants to compare satisfaction ratings for the same passengers between two service attributes: Seat comfort and Food and drink.

1. State the null and alternative hypotheses for testing whether there is a difference in satisfaction between seat comfort and food/drink ratings in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Create the difference variable. Also, check and interpret the assumption.
3. Run the appropriate test and report: mean difference, test statistic, df, p-value, and 95% CI.
4. Interpret the results. Is there a significant difference in satisfaction between these two service aspects?

QUESTION TWO (Group #2)

A restaurant chain is evaluating customer spending patterns. They want to test claims about average tips, compare spending between smokers and non-smokers, and assess whether spending differs between weekdays.

Use the dataset below to perform the following statistical tasks:

Dataset: Tips – Use:

- total_bill (spending amount)
- tip (tip amount)
- smoker (Yes/No)
- day (Thursday, Friday, Saturday, Sunday)
- time (Lunch/Dinner)
- sex (Male/Female)

Dataset link: https://raw.githubusercontent.com/mwaskom/seaborn-data/master/tips.csv

Part A: The restaurant claims that the average tip is \$3.00. The manager suspects the average tip is different.

1. State the null and alternative hypotheses in mathematical notation and in plain English. Indicate whether the test is one-tailed or two-tailed and why.

2. Perform the appropriate t-test in JASP/Jamovi to test whether the mean tip differs from \$3.00. Report: test statistic, df, p-value, and 95% CI.

3. Interpret the results and state your conclusion.

Part B: The manager wants to know whether smokers and non-smokers differ in their total spending (total_bill).

1. State the null and alternative hypotheses for testing whether there is a difference in total bill between smokers and non-smokers in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed and why.

2. Check and interpret the assumptions

3. Run the appropriate t-test and report the test statistic, df, p-value, and 95% CI for the difference.

4. Interpret the results and provide a recommendation.

Part C: The restaurant wants to compare spending on Thursday vs Friday for the same customers who visited both days. (Create a subset of customers with records for both days.)

1. State the null and alternative hypotheses for testing whether spending differs between Thursday and Friday in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed and why.

2. Create the difference variable. Also, check and interpret the assumption.

3. Run the appropriate t-test and report: mean difference, test statistic, df, p-value, and 95% CI.

4. Interpret the results. Is there a significant difference in spending between the two days?

QUESTION THREE (Group #3)

A company is evaluating the effectiveness of a new employee training programme. They have employee performance data from different departments. The HR director wants to know whether the training significantly improved scores.

Use the dataset below to perform the following statistical tasks:

Dataset: Employee Performance Dataset – use:

- PerformanceRating (performance score)
- Department (IT, Sales, HR, Support, e.t..c)
- TrainingHours (hours of training received)
- YearsAtCompany (years of experience)

Dataset link: https://www.kaggle.com/datasets/sayeeduddin/employee-performance-data-set

Part A: The industry average performance rating is 3.0. The HR director wants to know whether the company's employees differ from this benchmark.

1. State the null and alternative hypotheses in mathematical notation plain English. Indicate whether the test is one-tailed or two-tailed and why.

2. Perform the appropriate t-test in JASP/Jamovi to test whether the mean performance rating differs from 3.0. Report: test statistic, df, p-value, and 95% CI

3. Interpret the results and state your conclusion.

Part B: The company wants to compare performance ratings between two departments (e.g., Sales vs IT).

1. State the null and alternative hypotheses for testing whether there is a difference in performance ratings between the two departments in mathematical notation and in plain English. Indicate whether the test is one-tailed or two-tailed and why.

2. Check and interpret the assumptions

3. Run the appropriate test and report the test statistic, df, p-value, and 95% CI for the difference.

4. Interpret the results and provide a recommendation.

Part C: The company wants to know whether employees who received more training hours (e.g., > 20 hours) have higher performance ratings compared to when they had fewer training hours. (Create two groups based on training hours.)

1. State the null and alternative hypotheses for testing whether performance ratings differ between employees with high vs low training hours in mathematical notation and in plain English. Indicate whether the test is one-tailed or two-tailed and why.

2. Create the difference variable. Also, check and interpret the assumption.

3. Run the appropriate test and report: mean difference, test statistic, df, p-value, and 95% CI.

4. Interpret the results. Was the training programme effective?

QUESTION FOUR (Group #4)

A luxury jewellery retailer is evaluating diamond pricing strategies. They want to test various pricing hypotheses about diamond prices based on cut quality and carat weight.

Use the dataset below to perform the following statistical tasks:

Dataset: Diamonds Dataset – Use:
- price (diamond price in USD)
- cut (Fair, Good, Very Good, Premium, Ideal)
- carat (weight)
- color (D–J)
- clarity (I1, SI2, SI1, VS2, VS1, VVS2, VVS1, IF)

Dataset Link: https://huggingface.co/datasets/mstz/diamonds/raw/main/diamonds.csv

Part A: The retailer claims that the average diamond price is $4,000. The product manager suspects the average price is higher.

1. State the null and alternative hypotheses in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed and why.

2. Perform the appropriate test in JASP/Jamovi to test whether the mean price is greater than $4,000. Report: test statistic, df, p-value, and 95% CI.

3. Interpret the results and state your conclusion.

Part B: The retailer wants to compare prices between Ideal cut diamonds and Fair cut diamonds.

1. State the null and alternative hypotheses for testing whether there is a difference in price between Ideal and Fair cuts in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed and why.

2. Check and interpret the assumptions.

3. Run the appropriate test and report the test statistic, df, p-value, and 95% CI for the difference.

4. Interpret the results and provide a recommendation.

Part C: The retailer wants to compare prices of diamonds with carat weight near 1.0 carat vs near 1.5 carat. (Group carat values into "Near 1.0" and "Near 1.5" and create a subset with paired observations.)

1. State the null and alternative hypotheses for testing whether prices differ between 1.0 and 1.5 carat diamonds in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed and why.

2. Create the difference variable. Also, check and interpret the assumption.

3. Run the appropriate test and report: mean difference, test statistic, df, p-value, and 95% CI.

4. Interpret the results. Is there a significant difference in price

QUESTION FIVE (Group #5)

A retail chain is analysing sales performance. They want to evaluate customer spending, compare sales between customer segments, and assess the impact of discount strategies.

Use the dataset below to perform the following statistical tasks:

Dataset: Superstore Sales – Use:
- Sales (sales amount in USD)
- Customer Segment (Consumer, Corporate, Home Office, e.t.c)
- Category (Furniture, Office Supplies, Technology, e.t.c.)
- Discount (discount percentage)
- Profit (profit from sale)

Dataset Link: https://raw.githubusercontent.com/curran/data/gh-pages/superstoreSales/superstoreSales.csv

Part A: The chain claims that the average sales per order is $200. The manager suspects the average sales are lower.

1. State the null and alternative hypotheses in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed and why.

2. Perform the appropriate test in JASP/Jamovi to test whether the mean sales are less than $200. Report: test statistic, df, p-value, and 95% CI.

3. Interpret the results and state your conclusion.

Part B: The chain wants to compare average sales between Consumer and Corporate segments.

1. State the null and alternative hypotheses for testing whether there is a difference in sales between Consumer and Corporate segments in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed and why.

2. Check and interpret the assumptions.

3. Run the appropriate test and report the test statistic, df, p-value, and 95% CI for the difference.

4. Interpret the results and provide a recommendation.

Part C: The chain wants to compare sales before and after a discount campaign for the same product categories. *(Group by Category and compare Sales with Discount = 0 vs Discount > 0.)*

1. State the null and alternative hypotheses for testing whether sales increased after the discount campaign in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed and why.

2. Create the difference variable. Also, check and interpret the assumption.

3. Run the appropriate test and report: mean difference, test statistic, df, p-value, and 95% CI.

4. Interpret the results. Was the discount campaign effective?

QUESTION SIX (Group #6)

An automobile manufacturer is analysing fuel efficiency data. They want to evaluate MPG claims, compare vehicle types, and assess the impact of weight on fuel efficiency.

Use the dataset below to perform the following statistical tasks:

Dataset: Auto MPG – Use:

- mpg (miles per gallon)
- weight (vehicle weight in pounds)
- horsepower (engine horsepower)
- cylinders (number of cylinders)

Dataset Link: https://www.kaggle.com/datasets/zsinghrahulk/auto-mpg

Part A: The manufacturer claims that the average MPG is 30. The engineer suspects the average MPG is lower.

1. State the null and alternative hypotheses in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Perform the appropriate test in JASP/Jamovi to test whether the mean MPG is less than 30. Report: test statistic, df, p-value, and 95% CI.
3. Interpret the results and state your conclusion.

Part B: The manufacturer wants to compare MPG between 4-cylinder and 6-cylinder cars.

1. State the null and alternative hypotheses for testing whether there is a difference in MPG between 4-cylinder and 6-cylinder cars in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Check and interpret the assumptions.
3. Run the appropriate test and report the test statistic, df, p-value, and 95% CI for the difference.
4. Interpret the results and provide a recommendation.

Part C: The manufacturer wants to compare fuel efficiency between lighter and heavier vehicles within the same cylinder group. (Group by cylinders, then compare MPG for vehicles below vs above median weight.)

1. State the null and alternative hypotheses for testing whether MPG differs between lighter and heavier vehicles in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Create the difference variable. Also, check and interpret the assumption.
3. Run the appropriate test and report: mean difference, test statistic, df, p-value, and 95% CI.
4. Interpret the results. Is weight reduction likely to improve fuel efficiency?

QUESTION SEVEN (Group #7)

A company is evaluating employee productivity across different departments. They have collected performance data from employees in various departments. The HR director wants to know whether performance ratings differ by department and whether work-life balance affects performance.

Use the dataset below to perform the following statistical tasks:

Dataset: Employee Performance Dataset – Use:

- PerformanceRating (performance score 1–5)
- Department (IT, Sales, HR, Support, e.t.c.)
- WorkLifeBalanceScore (work-life balance score)
- YearsAtCompany (years of experience)

Dataset Link: https://www.kaggle.com/datasets/sayeeduddin/employee-performance-data-set

Part A: The industry benchmark for performance rating is 3.0. The HR director wants to know whether the company's employees meet this benchmark.

1. State the null and alternative hypotheses in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Perform the appropriate test in JASP/Jamovi to test whether the mean performance rating differs from 3.0. Report: test statistic, df, p-value, and 95% CI.
3. Interpret the results and state your conclusion.

Part B: The company wants to compare performance ratings between the Sales and IT departments.

1. State the null and alternative hypotheses for testing whether there is a difference in performance ratings between the two departments in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Check and interpret the assumptions.
3. Run the appropriate test and report the test statistic, df, p-value, and 95% CI for the difference.
4. Interpret the results and provide a recommendation.

Part C: The company wants to compare performance ratings between employees with high work-life balance scores vs low work-life balance scores within the same department. (Group by Department, then split on WorkLifeBalanceScore median.)

1. State the null and alternative hypotheses for testing whether performance ratings differ between high and low work-life balance employees in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Create the difference variable. Also, check and interpret the assumption.
3. Run the appropriate test and report: mean difference, test statistic, df, p-value, and 95%

CI.

4. Interpret the results. Is work-life balance associated with higher performance?

QUESTION EIGHT (Group #8)

A technology company operates a customer support centre that handles thousands of support tickets each month. They have collected data on support ticket resolution and customer satisfaction. The operations manager wants to:

a) Verify whether the average response time meets the company's target.

b) Compare satisfaction levels between different support channels.

c) Assess whether resolution time differs between tickets handled by experienced versus less experienced agents.

Use the dataset below to perform the following statistical tasks:

Dataset: Customer Support Ticket Satisfaction Analysis Dataset – Use:

- Response_Time (response time in minutes)
- Resolution_Time (resolution time in hours)
- Agent_Experience (years of experience)
- Support_Channel (Email, Chat, Phone)
- Customer_Satisfaction (Low, Medium, High)
- Ticket_Priority (Low, Medium, High, Urgent)
- Issue_Category (Billing, Technical, Account, Delivery, Other)

Dataset Link: https://www.kaggle.com/datasets/jayjoshi37/customer-support-ticket-satisfaction-analysis/data

Part A: The company claims that the average response time is 30 minutes. The operations manager believes the average response time is higher than 30 minutes.

1. State the null and alternative hypotheses in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Perform the appropriate test in JASP/Jamovi to test whether the mean response time is greater than 30 minutes. Report: test statistic, df, p-value, and 95% CI.
3. Interpret the results and state your conclusion.

Part B: The company wants to compare resolution times between tickets handled via Email vs Phone support channels.

1. State the null and alternative hypotheses for testing whether there is a difference in resolution time between Email and Phone support channels in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Check and interpret the assumptions.
3. Run the appropriate test and report the test statistic, df, p-value, and 95% CI for the difference.
4. Interpret the results and provide a recommendation.

Part C: The company wants to compare response time and resolution time for the same tickets to understand whether faster responses lead to faster resolutions.

1. State the null and alternative hypotheses for testing whether there is a difference between response time and resolution time in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Create the difference variable. Also, check and interpret the assumption.
3. Run the appropriate test and report: mean difference, test statistic, df, p-value, and 95% CI.
4. Interpret the results. Is there a significant difference between response time and resolution time?

QUESTION NINE (Group #9)

A marketing team is evaluating campaign performance and customer satisfaction. They have data from different departments and customer segments. They want to know whether performance ratings differ by department and whether customer satisfaction affects performance.

Use the dataset below to perform the following statistical tasks:

Dataset: Employee Performance Dataset – Use:

- PerformanceRating (performance score 1–5)
- CustomerSatisfaction (customer satisfaction score 0–10)
- Department (IT, Sales, HR, Support, e.t.c.)
- TrainingHours (hours of training received)

Dataset Link: https://www.kaggle.com/datasets/sayeeduddin/employee-performance-data-set

Part A: The marketing team's target performance rating is 3.5. The manager wants to know whether the team meets this target.

1. State the null and alternative hypotheses in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Perform the appropriate test in JASP/Jamovi to test whether the mean performance rating differs from 3.5.. Report: test statistic, df, p-value, and 95% CI.
3. Interpret the results and state your conclusion.

Part B: The team wants to compare performance ratings between the Sales and HR departments.

1. State the null and alternative hypotheses for testing whether there is a difference in performance ratings between the two departments in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Check and interpret the assumptions.
3. Run the appropriate test and report the test statistic, df, p-value, and 95% CI for the difference.
4. Interpret the results and provide a recommendation.

Part C: The team wants to compare performance ratings between employees with high customer satisfaction scores vs low customer satisfaction scores within the same department. (Group by Department, then split on CustomerSatisfaction median.)

1. State the null and alternative hypotheses for testing whether performance ratings differ between high and low customer satisfaction employees in mathematical notation and plain English. Indicate whether the test is one-tailed or two-tailed, and explain why.
2. Create the difference variable. Also, check and interpret the assumption.
3. Run the appropriate test and report: mean difference, test statistic, df, p-value, and 95% CI.
4. Interpret the results. Is customer satisfaction associated with higher employee performance?